



Documentation: methods, orthography, translation, variation

Ling 98A Junior Tutorial [Fall 2023]

Instructor: Yvette Yi-Chi Wu



**we will discuss some
methodological
issues relevant to
language
documentation:**

1. the people involved
2. data collection
and processing
3. writing systems,
grammar, dictionaries
4. translation
5. variation and
standardization

issue #1: the people involved



possible actors

- community members
- linguists
- educators
- families
- local government
- language institutions
- other academics/specialists:
 - anthropologists
 - biologist/ecologists
 - artists/musicians
 - historians



Mashpee Wampanoag tribal educators



Steve Aoki (linguist) with Nez Perce scholars
Angel Sobotta (left) and Beth Piatote (right)



National Breath of Life Institute 2013

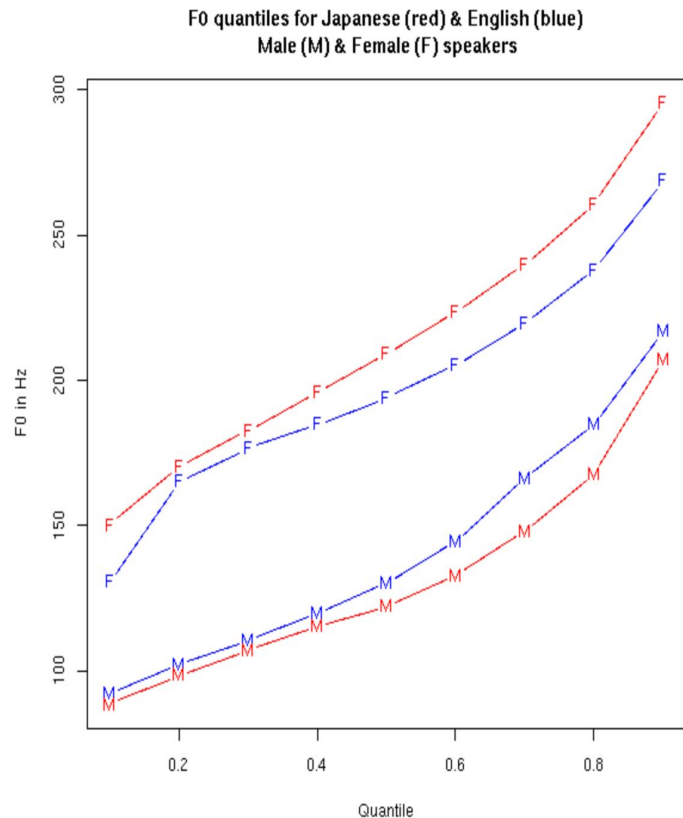


master-apprentice program through the AICLS



community engagement

- different generations (especially elders)
- different genders (men, women, and others)
- speakers of different dialects *and* registers
- fluent and non-fluent speakers
- community members with different roles in the community
- what is the importance of including different community members?



differences in pitch across; Liberman (2007)

CHAPTER FOUR

RITUAL LANGUAGES, SPECIAL REGISTERS AND SPEECH DECORUM IN AUSTRONESIAN LANGUAGES

James J. Fox

1 INTRODUCTION

Ritual languages, special registers and speech decorum reflect the social and cultural features of the societies in which they occur. They thus constitute linguistic usages that are significantly shaped by extra-linguistic factors. Yet the patterning of these usages in Austronesian languages shows remarkable similarities across a wide range of diverse societies. Such similarities point to the operation of some of the basic features of

R. M. W. DIXON

Edible Gender, Mother-in-Law Style, & Other Grammatical Wonders

*Studies in Dyirbal, Yidiñ,
& Warrgamay*

OXFORD
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involvement of outsiders

- many communities prefer not to have outsiders involved in language reclamation efforts, and prefer to have their linguistic knowledge kept safe within their community
- this includes not wishing to teach outsiders the language
- what are the reasons behind this?



potential roles of an outsider linguist

- **documentation**: record language and grammar
- **development of grammar, orthography, dictionaries**: linguistic expertise and formal analysis can guide and facilitate the production of these materials
- **research**: typological/historical research can help when there are few existing records; can also bring exposure and seen as a status symbol
- **teaching and training**: teaching the language and training native linguists
- **administration and program development**: linguists may have better access or knowledge of administrative processes, funding sources, etc.

issue #2: data collection and processing



language data collection

- what are the pros and cons of each data gathering method?
 - oral histories, personal narratives, stories and songs
 - structured elicitation, pictures/storyboards, sociolinguistic interviews
 - conversations (one-on-one and large group setting)
 - naturalistic observation, traditional practices (rituals, etc.)
- describe the consultants and their background
- describe your own background and personal (linguistic) biases



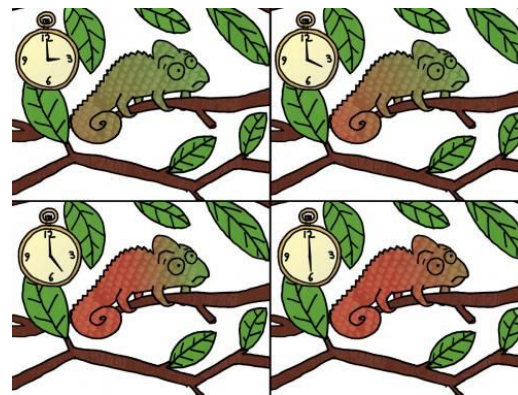
PhD candidate Gladys Camacho Rios working with speakers of her native language Quechua



Family Problems Picture Task



linguists working in the Lower Fungom community



storyboard for gathering data about time

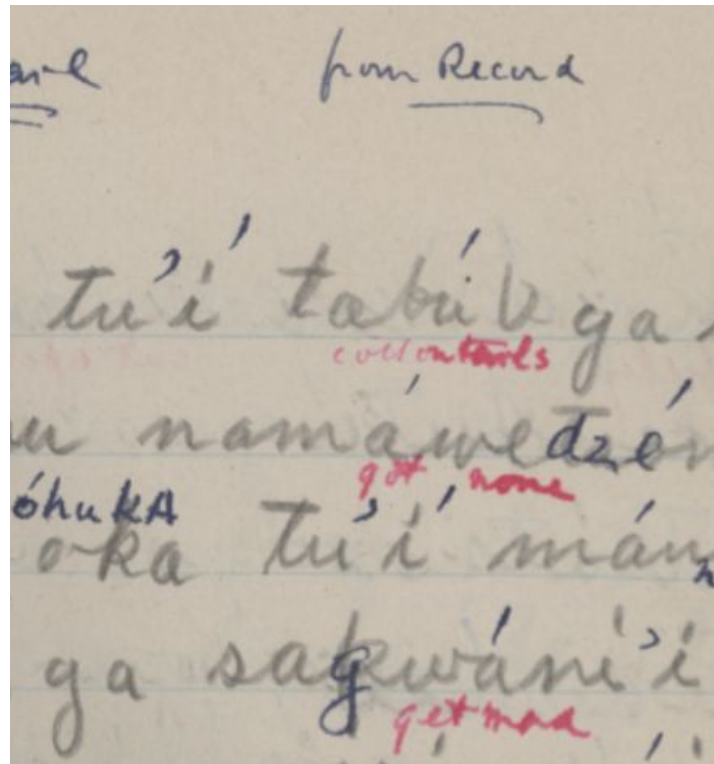


some resources to aid documentation

- Family Problems Picture Task:
<https://scopicproject.wordpress.com/>
- Scope Fieldwork Project:
https://www.eva.mpg.de/lingua/tools-at-lingboard/material_scope_fieldwork_project.php
- Totem Field Storyboards:
<https://www.totemfieldstoryboards.org/>
- The Pear Story:
<https://www.youtube.com/watch?v=bRNSTxTpG7U>

archival material

- language data can also come from archival material and texts
 - past notes from linguists
 - old recordings or speakers
 - historical texts (e.g. bible)
 - government documents
- must be interpreted with caution—lack of context and possible errors



[California Language Archives \(CLA\)](#)



data organization and participant information

- information to include with transcription and audio/video files:
 - date, duration, location, setting (home, school, event, etc.)
 - participants, researcher; subject language(s), meta language(s)
 - mode (elicitation, narrative, etc.) and topic (linguistic or cultural)
 - dialects, orthography used, and other relevant information
- please be extremely **sensitive to people's consent** to have their information recorded, to sensitive information in the data collected, etc.



language data processing

- ELAN:
 - create time-aligned transcriptions of audio
- FLEx:
 - create a lexicon, add audio, pictures, and video to entries
 - transcribe stories, texts, and sentences with interlinear glosses
 - analysis of morphemes, including allomorphy, boundedness, etc.
- Praat:
 - annotate audio phonetically with text grids
 - produces spectrograms to calculate formants, pitch tracks, etc.

Sena 3 - FieldWorks Language Explorer

File Edit View Data Insert Format Tools Parser Window Help

Lexicon Entries Entry Show Hidden Fields

Headword Lexeme Form Glosses (Eng) Glosses (Por)

nkura 1) make grow 2) raise a child crisar crisanga 3) magnify magnificar (religiosa)

Lexeme Form Sen nkuz

Morph Type seot

Citation Form Sen nkura

Components

Note Eng Por

Singular Plural

Insert Example Insert Sense

Eng make grow

Por faster crescer

Por make grow

Definition

Grammatical Info. Verb

Example

Semantic Domains

Lexical Relations

Parsing Note

Eng raise

Por crisar

Eng raise a child

Por crisar crisanga

Definition

Grammatical Info. Verb

Example

Semantic Domains

Sorted by Lexeme Power Filtered 116/268

Headword	Lexeme Form	Glosses (Eng)	Glosses (Por)
nkhaba	nkhaba	not 1	não 1
nkhodolo	nkhodolo	back of head	nuca
nkhomole	nkhomole	cliff	precipício
nkolo	nkolo	hemorrhoids	hemorróides
nkono-nkono	nkono-nkono	right side	a direita
nkucama	nkucama	day after tomorrow	depós de amanhã
nkudza	nkudz	become great	engrandecer
nkura	nkuz	make grow raise magnify	faster crescer crescer magnificar
noswe	noswe	seven	sete
nos	nos	sharpen	afiar
nsanga	nsanga	fast	depressa
nsanga-nsanga	nsanga-nsanga	very fast	muito depressa
nsua	nsua	island	ilha
nsunga	nsunga	yeast liquid	fermento
nsuzi	nsuzi	grave	molho
ntali	ntali	far in	muito dentro
nthalaunda	nthalaunda	area	area
nthenda	nthenda	a sickness	uma doença
nthenga	nthenga	feather	pena de ave

FLEX dictionary entries

ELAN - elan-example1.xlf

File Edit Annotation Tier Type Search View Options Window Help

Menu bar

Grid Text Subtitles Lexicon Audio Recognizer Video Recognizer Metadata Controls

Recognizers and viewers

Media controls

Video viewer

Volume: 100

Rate: 100

00:00:11:01 Selection: 00:00:00:000 - 00:00:00:000 0

Annotation density viewer

Waveform viewer

Tiers

Timeline viewer

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and you follow then the sig that's the oren

the you foli sign kleaf

con prov adv a n n de v a n n

and ju folao den be sam kle f

then you follow the sign

the you foli sign kleaf

adv ju folao be sam kle f

arato stroke hold preparation hol stroke hold preparato stroke hold preparation hold

Going to

Going alo

A roundabout

G

ELAN timed transcription with audio and video



storing language data

- the language data collected should be stored in a manner that is accessible to the community and (if permission granted) to the public
- some open access language archives:
 - Open Language Archives Community (OLAC)
 - Endangered Languages Archive (ELAR)
 - Digital Endangered Languages and Musics Archives Network (DELAMAN)
 - YouTube and other social media platforms
- what factors should we keep in mind in making language data accessible?



OLAC: Open Language Archives Community

[HOME](#) | [DOCUMENTS](#) | [ABOUT](#) | [ARCHIVES](#)
[NEWS](#) | [ORGANIZATION](#) | [TOOLS](#) | [SERVICES](#)

SEARCH THIS SITE:

OLAC Mission

OLAC, the Open Language Archives Community, is an international partnership of institutions and individuals who are creating a worldwide virtual library of language resources by: (i) developing consensus on best current practice for the digital archiving of language resources, and (ii) developing a network of interoperating repositories and services for housing and accessing such resources.

News

OLAC Joins the Linguistic Linked Open Data Cloud: The OLAC system has now been integrated with LLOD... [More...](#)

New OLAC Page Listing Archive Submission Policies: As a service to linguists who are in search of an archive that could receive a deposit... [More...](#)

New OLAC Search Service: In December 2010, to mark our 10th anniversary, OLAC announces a new search service... [More...](#)

[More news ...](#)

Documents

[OLAC Standards](#) - specify how OLAC operates

[Recommendations](#) - express consensus of OLAC members regarding language resource archiving

Find Language Resources

Use OLAC's search engine at: <http://search.language-archives.org/>



Archive:

Region: [Africa](#) [Americas](#) [Asia](#) [Europe](#) [Pacific](#)

OLAC Coverage

OLAC Archives contain over 300,000 records, covering resources in half of the world's living languages. [More statistics on coverage.](#)

Join the OLAC Community



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issue #3: writing systems, grammar, dictionaries

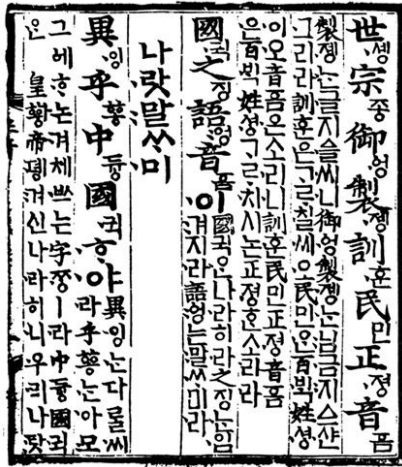


types of scripts

- **alphabet**: characters represent phonemes, e.g. Latin alphabet
- **syllabary**: characters represent syllables, e.g. Japanese katakana/hirigana
- **logogram**: characters represent words, e.g. Egyptian hieroglyphs
- **abjad**: characters represent consonants (vowels inferred), e.g. Arabic abjad
- **abugida**: characters represent CV sequences, e.g. Javanese aksara jawa
- what are the pros and cons of each type of script?



Mayan script (logosyllabic)



Hangul (Korean) script (syllabic alphabet)

	C	Ca	Cu	Ci	Ce	Co	Cā	Cū	Cē	Cō	C̄	C̄
?	3	3	3	3	3	3	3	3	3	3	3	3
b	2	2	2	2	2	2	2	2	2	2	2	2
d	3	3	3	3	3	3	3	3	3	3	3	3
d'	4	4	4	4	4	4	4	4	4	4	4	4
h'	3	3	3	3	3	3	3	3	3	3	3	3
w	2	2	2	2	2	2	2	2	2	2	2	2
z	2	2	2	2	2	2	2	2	2	2	2	2
h²	2	2	2	2	2	2	2	2	2	2	2	2
t	2	2	2	2	2	2	2	2	2	2	2	2
Y	2	2	2	2	2	2	2	2	2	2	2	2
K	2	2	2	2	2	2	2	2	2	2	2	2
l	2	2	2	2	2	2	2	2	2	2	2	2
m	2	2	2	2	2	2	2	2	2	2	2	2
n	2	2	2	2	2	2	2	2	2	2	2	2
s¹	2	2	2	2	2	2	2	2	2	2	2	2
F	2	2	2	2	2	2	2	2	2	2	2	2
S²	2	2	2	2	2	2	2	2	2	2	2	2
K	2	2	2	2	2	2	2	2	2	2	2	2
r	2	2	2	2	2	2	2	2	2	2	2	2
S	2	2	2	2	2	2	2	2	2	2	2	2
t	2	2	2	2	2	2	2	2	2	2	2	2
x	2	2	2	2	2	2	2	2	2	2	2	2
g	2	2	2	2	2	2	2	2	2	2	2	2
ç	2	2	2	2	2	2	2	2	2	2	2	2
ñ	2	2	2	2	2	2	2	2	2	2	2	2
č	2	2	2	2	2	2	2	2	2	2	2	2
ř	2	2	2	2	2	2	2	2	2	2	2	2

Sapolo (Oromo) script (abugida)



MS 577
Syriac script, Mt. Sinai, Egypt, ca. 11th c.

Syriac script (abjad)



orthography

- the Latin/Roman alphabet is by far the most popular script chosen
- what are factors in adopting an existing script vs. developing a new script?
 - character to sound correspondence
 - phonological structure of a language
 - suprasegmental information (e.g. tone, nasality, creakiness, etc.)
 - accessibility and literacy
 - time and resources needed to develop a new script

Figure 7. code-internal differences between LPO's and EPO's

Linguistic Practical Orthographies (LPO)	English-based Practical Orthographies (EPO)
one symbol, one sound	various sounds per symbol and vice versa
modelled on IPA/Americanist orthographic conventions	modelled on English words that have same sound as a given syllable in the language
no syllable breaks indicated	indicates syllable breaks
represents phonemic system of native speakers	tendency toward subphonemic representation or English-influenced phonemic system
preserves suprasegmental features such as vowel length, glottalization	tends to leave out suprasegmentals



phonetic detail

- how much phonetic detail is relevant (i.e. distinctive)?
- certain details can be left out of the orthography if it is predictable
- Seediq: vowels three or more syllables from the end are omitted
 - *tgdaya* ‘up the hill’, *mnmekan* ‘want to eat’, *qbheni* ‘bird’
- English: aspiration is not encoded in the writing system
 - *table* [t^heɪbəl] vs. *stable* [steɪbəl] vs. *tablet* [t^hæblət]



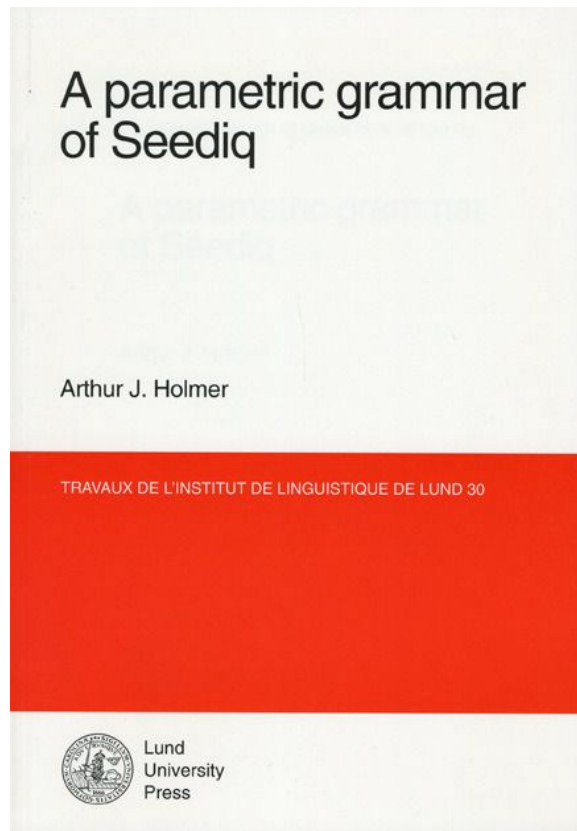
mismatches

- when a script is adopted for a language, **there are often mismatches** that come with the script not being designed for that language
 - **Latin alphabet for Javanese**: diagraphs *dh*, *kh*, *ng*, *ny*, *sy*, and *th*
 - **Latin alphabet for Indonesian**: no distinction between *e* for [e] and [ə]
 - **Latin alphabet for Seediq**: no character for [ʔ]
- borrowing of English into Japanese: McDonald's → マクドナルド

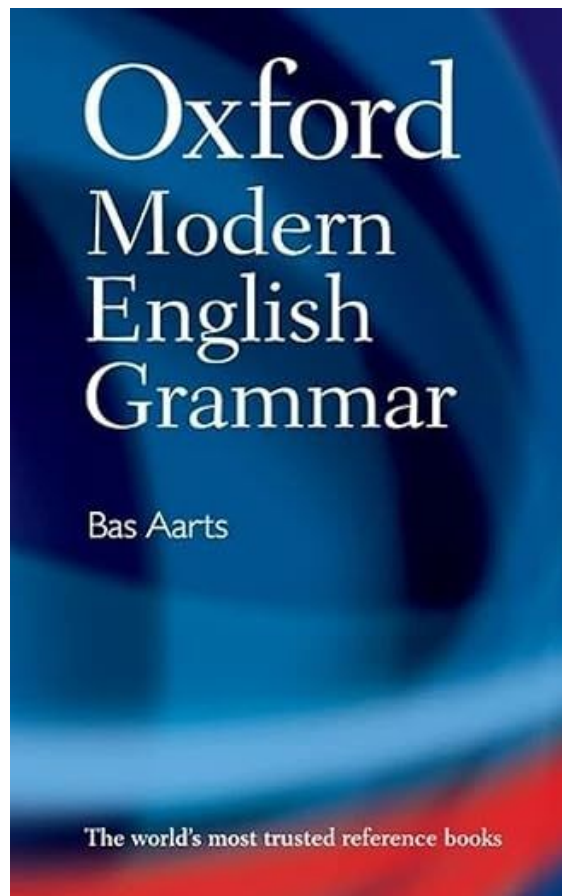


grammars and dictionaries

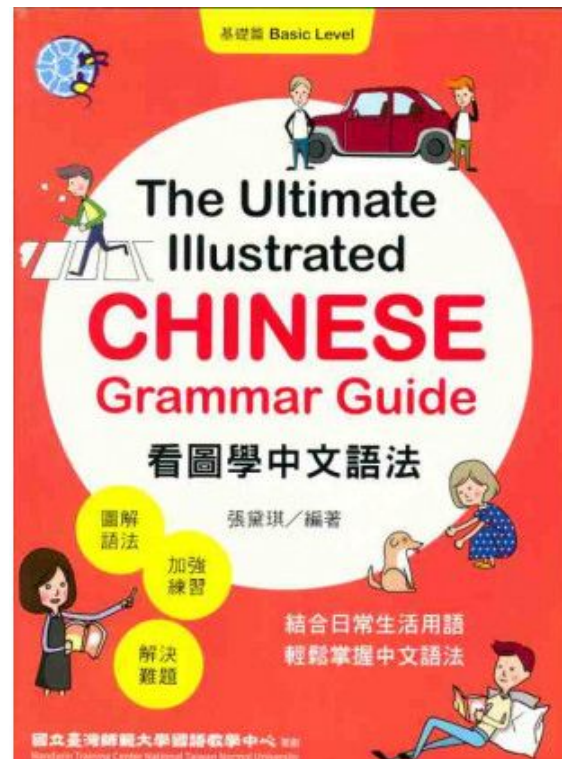
- considerations for **target audience**
 - academic linguists, language learners, or both
 - amount of linguistics jargon, framework, and presentation
- considerations for **language variation and dialects**
 - specify which dialect(s) the grammar/dictionary is based on
 - ideally, you can credit individual speakers for each data point
- **how much data is enough to start writing a grammar/dictionary?**



Seediq grammar for linguists



English grammar for linguists
and advanced learners



Mandarin grammar for learners



morphological complexity

- how should a dictionary be constructed for languages with high morphological complexity (e.g. many conjugated forms)?
- verbs can have upwards of hundred of forms in a single language
- can (ir)regularities in morphology can be represented in a dictionary?
- Spanish: verbs have irregular forms depending on tense
 - tener 'to have' is *tengo* in 1SG.PRES but *tuve* in 1SG.PST

issue #4: translation



metalanguage

- a **metalanguage** is a language used to describe/teach/talk about another
- potential biases introduced by a metalanguage:
 - what is documented and translated
 - influence of grammatical structure
- difficulty conveying culturally specific uses, terms, proverbs, slangs



what is equivalence in translation?

- sometimes the metalanguage does not encode the same grammatical information, and translations can either omit or add information:
 - tense, aspect, mood (TAM) marking
 - information structure, word order, pronoun dropping
 - active/passive, verbal/nominal/adjectival stems
 - definiteness, gender, number, person, honorifics
- how do we avoid the linguistic biases and pitfalls of translation?

voice alternations (Paiwan; Ferrell 1979)

- a. q<**m**>alup [a caucau] tua vavuy i gadu tua vulu
 <AV>hunt NOM man OBL pig LOC mountain OBL spear
- b. qalup-**en** nua caucau [a vavuy] i gadu tua vulu
 hunt-PV ERG man NOM pig LOC mountain OBL spear
- c. qalup-**an** nua caucau tua vavuy [a gadu] tua vulu
 hunt-LV ERG man OBL pig NOM mountain OBL spear
- d. **si**-qalup nua caucau tua vavuy i gadu [a vulu]
 IV-hunt ERG man OBL pig LOC mountain NOM spear
 'The man hunts wild pigs in the mountains with a spear.'

underspecified number (Mandarin Chinese)

- a. *Wo mai le shu.*
 1.SG buy ASP book
 'I bought a/some book(s).'
- b. *Ni kanjian le xuesheng.*
 2.SG see ASP student
 'You saw a/some student(s).'

null pronouns (Passamaquoddy; Mitchell 1921)

- Pol '-koti-wsikiya-l eli-milomahtu-li-t
first 3-Fut-punish-Obv IC.thus-be.mischievous-ObvS-3Conj
'First, he will punish him for the many ways he has behaved'

Icibemba tense and aspect marking on verbs

Tense/aspect	Prefix	Stem	Notes
Present continuative	le	unmodified	Used for verbs of action to indicate a progressive action; can be used adjectivally, and even as an immediate future (especially when used with verbs of state).
Present definite	a	unmodified	Used for verbs of state, and intransitive verbs
Stative, necessary habitual	Ordinary form; no prefix	unmodified	Indicates a state that is always true, or a habitual action; connotes necessity of the action
Voluntary habitual	la	unmodified	For habitual actions, connoting that they are voluntary, or may otherwise not be necessary; used only for verbs of state; can be used in subordinate clauses
Immediate past	a	unmodified	For actions occurring 'today', or 'nowadays'
Recent, completed or historic past	na- (prefix)	unmodified	For actions completed today; can be used in a historic sense, but does not emphasise a distant past
Specific past	(no prefix)	modified	Also used for actions done 'today' or 'nowadays', but emphasising more distance in time than the previous two; used in relative clauses
Simple past of yesterday (1)	ali	unmodified	The stress must not be on the 'a' of the prefix
Simple past of yesterday (2)	a	modified	The stress must not be on the prefix
Remote past simple	á	modified	
Remote past of duration	ali	modified	For remote actions whose effects are still present, or for states established in the remote past which have not changed
Remote past definite, past perfect	áli	unmodified	A remote past further distant than some other relevant time
Past imperfect or habitual	ale	unmodified	Used for actions once, but no longer, habitual
Immediate future	ala	unmodified	
Immediate progressive or habitual future	akula	unmodified	
Proximal future	le	unmodified	For future states and conditions to be established shortly, but not necessarily immediately
Future indefinite	ka	unmodified	For actions and states at some unknown time in the future
Future indefinite habitual	kala	unmodified	For habitual actions and states at some unknown time in the future



what are some strategies to elicit the following?

- tense, mood, aspect e.g. (multiple) past/present/future
- (in)definiteness, specificity e.g. the/a/that cat
- person, number, gender e.g. singular/dual/plural, count vs. mass
- when documenting a language, it is **always important to set up (culturally relevant) contexts** that will help to disambiguate the meaning of a sentence

issue #5: variation and standardization



varieties

- also called **sociolects, idiolects, registers, dialects, languages, styles**, etc.
- varieties can differ on multiple grounds:
 - historical divergence, contact with other languages
 - social setting (e.g. formality, ritual languages, secret languages, etc.)
 - within an individual vs. across individuals or groups
 - influence from writing systems and regulatory bodies
- there can sometimes be tensions between speakers of different varieties



language change

- documentation can be complicated when a language is rapidly changing
 - due to language shift and/or attrition, or new speakers
- need for new vocabulary to accommodate to changing lifestyle
 - communities can disagree on the extent of allowing loan words
- what complexities does language change bring to documentation?
- how much should language pedagogy adapt to changing language?



standardization

- a process of determining an official and internally consistent version of a language, typically **decided by an institution and used in education**
- this includes aspects such as:
 - grammar, vocabulary, pronunciation, orthography
- sometimes unintended standardization happens when limited resources entail that documentation is limited to certain dialects or varieties
- **what are the pros and cons of standardizing a language?**

Rat für deutsche Rechtschreibung

Council for German Orthography



Academy of the Arabic Language (Egypt)



Language Development and Fostering Agency
(Indonesia)



Maori Language Commission



National Swahili Council (Tanzania)

aspects to consider

- education and pedagogy
 - spoken vs. written communication
 - identity and heritage
 - creativity and expressiveness
 - community building and unity
- can you implement standardization while valuing variation/diversity?

